

Mohsen Taheri Shalmani

Email: MohsenTaheriShalmani@gmail.com

Web: <https://orcid.org/0000-0003-4044-8507>

Tel: +4741005164

EDUCATION

(2020 – 2024) Ph.D. in IT, Mathematics and Physics, University of Stavanger.

(2018 – 2020) MSc in Mathematics and Physics, University of Stavanger.

(2007 – 2009) MSc in Applied Mathematics, Azad University, Science and Research Branch.

(2002 – 2006) BSc in Mathematics, Azad University, Tehran Central Branch.

WORK EXPERIENCE

(2025 – Pres.) AI researcher in [EAST Project](#), Kristiania University of Applied Sciences, Oslo.

(2025 – Pres.) Lecturer at the University of Stavanger (20% position).

(2020 – 2024) Doctoral researcher and a member of [BMDLab](#), University of Stavanger.

(2015 – 2018) Math and computer teacher, Mojtama Fani, Tehran.

(2010 – 2015) Bank clerk and IT consultant, Karafarin Bank, Tehran.

(2009 – 2010) Freelance math and computer teacher, Tehran.

(2006 – 2007) Math teacher, Sadr High School, Tehran.

SKILLS

Statistical Modeling and Simulation | R, Python, and MATLAB programming | Mathematical Modeling | AI and ML | Deep Learning | Image Processing and LLM via TensorFlow and Transformers | Big Data | PySpark | Data Mining | Differential Equations | Numerical Analysis | Time Series | Signal Processing | Object Modeling | 3D Slicer | Functional Data Analysis | Language Processing | Computational Geometry | Optimization | Neuroimaging | Object-Oriented Programming | Java, Kotlin, C++, C# | System Testing | Containerization | Docker | Database Validation | Web Applications | React | JavaScript | Power BI | Excel | Cloud Computing | Database Programming | Azure | SQL |
GitHub: <https://github.com/MohsenTaheriShalmani>

Open-Source Contributions

- [EvoMaster](#) – Contributing to advancing the EvoMaster open-source automated testing tool by equipping it with AI technologies to enhance API testing, improving reliability, security, and automated generation of test cases for web and enterprise systems.
- [Elliptical Tubes](#) – Development of the [ETRep R package](#) and [Elliptical Tube Interactive App](#) designed for statistical shape analysis of tubular structures, enabling object modeling, simulation, and mean shape computation of elliptical tubes for applications in biomedical and geometric data analysis.
- [SlicerSALT](#) – Contributor in developing hypothesis testing procedures for shape models within the SlicerSALT open-source platform, which provides advanced tools for statistical shape analysis, visualization, and morphometric measurements for biomedical research.

LANGUAGES

English (Fluent) | Norwegian (Intermediate) | Persian (Native).

SUPERVISION

Co-supervised master's thesis: “*Statistical Shape Analysis of Caudate Nucleus in Parkinson's Disease*” (2025) by Thomas Eikemo Tjensvold, University of Stavanger. <https://hdl.handle.net/11250/3209793>

PUBLICATIONS

- Taheri Shalmani, M., Arcuri, A., Galeotti, J.P., Golmohammadi, A. “Online Machine Learning Prediction of Failed Calls to Enhance REST API Fuzzing” (2026) (Under review)
- Arcuri, A., Zhang, M., Seran, S., Galeotti, J.P., Golmohammadi, A., Garrett, P., Sahin, O., Castagna, F., Kertusha, Taheri Shalmani, M., I. and Susilo, F.F., 2026. “EvoMaster at REST League 2026 Tool Competition”. In 19th Search-Based and Fuzz Testing Workshop (SBFT’26).
- Pizer, S. M., Liu, Z., Zhao, J., Tapp-Hughes, N., Damon, J., Zhang, M., Marron, J. S., Taheri Shalmani, M., & Vicory, J. (2025). “Interior object geometry via fitted frames”. Journal of Mathematical Imaging and Vision. <https://doi.org/10.1007/s10851-025-01266-6>
- Taheri Shalmani, Mohsen (2025) “Deriving Spines of Tubular Objects via Deep Learning. (Preprint) <https://doi.org/10.13140/RG.2.2.14238.45126>
- Pizer, S. M., Marron, J., Taheri, M., & Vicory, J. “Object Correspondence for Shape Statistics”. (2024–25) Interactions of Statistics and Geometry (ISAG) II.
- Taheri Shalmani, Mohsen, Stephen M. Pizer, and Jörn Schulz. (2025) [R package] ETRep “Analysis of Elliptical Tubes Under the Relative Curvature Condition”. The Comprehensive R Archive Network (CRAN). <https://doi.org/10.32614/CRAN.package.ETRep>
- Taheri Shalmani, Mohsen. “Shape Statistics via Skeletal Structures.” (2024) Ph.D. Thesis, University of Stavanger, Norway. <https://doi.org/10.13140/RG.2.2.34500.23685>
- Taheri Shalmani, M., Pizer, S. M., & Schulz, J. (2024). “The Mean Shape under the Relative Curvature Condition”. Journal of Computational and Graphical Statistics. <https://doi.org/10.1080/10618600.2025.2535600>
- Taheri, Mohsen, Stephen M. Pizer, and Jörn Schulz. (2023) “Fitting the Discrete Swept Skeletal Representation to Slabular Objects.” (Under review) <https://doi.org/10.48550/arXiv.2409.04079>
- Taheri, Mohsen, and Jörn Schulz. (2022) “Statistical Analysis of Locally Parameterized Shapes.” Journal of Computational and Graphical Statistics 32, no. 2: 658-670. <https://www.tandfonline.com/doi/full/10.1080/10618600.2022.2116445>
- Liu, Z., Schulz, J., Taheri, M., Styner, M., Damon, J., Pizer, S., & Marron, J. S. (2023). “Analysis of joint shape variation from multi-object complexes”. Journal of Mathematical Imaging and Vision, 65(3). <https://link.springer.com/article/10.1007/s10851-022-01136-5>
- Pizer, Stephen M., J. S. Marron, James N. Damon, Jared Vicory, Akash Krishna, Zhiyuan Liu, and Mohsen Taheri. (2022) “Skeletons, object shape, statistics.” Frontiers in Computer Science 4: 842637. <https://www.frontiersin.org/articles/10.3389/fcomp.2022.842637/full>
- Taheri Shalmani, Mohsen. (2020) “Statistical shape analysis of brain structures.” Master's thesis, University of Stavanger. <https://doi.org/10.13140/RG.2.2.12270.42567>

REFERENCES

- | | |
|---|---|
| • Dr. Jörn Schulz (UiS) | Email: jorn.schulz@uis.no |
| • Dr. Jan Terje Kvaløy (UiS) | Email: jan.t.kvaloy@uis.no |
| • Dr. Stephen M. Pizer (UNC) | Email: smp@cs.unc.edu |
| • Dr. Andrea Arcuri (Kristiania University) | Email: andrea.arcuri@kristiania.no |